HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed **Python**-based AI agents using **LangGraph** and **tool calling** capabilities to automate multi-step financial reconciliation workflows, reducing manual intervention by an estimated **40%**.
- Designed and orchestrated **multi-agent systems (MAS)** for autonomous operation of a distributed data ingestion pipeline, enabling seamless **Agent-to-Agent (A2A)** communication for task delegation and error handling.
- Implemented and optimized **LLM inference pipelines** for high-throughput, low-latency serving of analytical models, achieving a p95 latency of **180ms** using **FastAPI** and **vLLM**.
- Contributed to the design and implementation of **AI/ML solutions** by collaborating with cross-functional product teams to translate business requirements into technical specifications for new agentic functionalities.
- Monitored the performance of deployed **machine learning models** and agentic workflows, collecting key metrics and analyzing data to fine-tune algorithms and enhance system accuracy, leading to a **15%** reduction in false positives for anomaly detection.
- Authored comprehensive technical documentation and presented strategic findings on **AI advancements** and their potential impact to senior stakeholders, influencing roadmap decisions for future product development.
- Containerized AI agent services using **Docker** and deployed them to an **AWS EKS** cluster, managing configuration with **Helm** and ensuring scalable, resilient operation.

SKILLS

AI/ML – Machine Learning, Deep Learning, Reinforcement Learning, AI Agents, Multi-Agent Systems (MAS), Autonomous Agentic Workflows, Tool Calling, Agent-to-Agent (A2A) Communication, LLM Inference Pipelines, Model Optimization, Algorithm Tuning, Performance Metrics, AI/ML Solutions Design

Programming Languages – Python, Java, Go

Frameworks & Libraries – LangGraph, AutoGen, CrewAI, Semantic Kernel, vLLM, SGLang, FastAPI, Flask, Pandas, NumPy, Scikit-learn

Cloud & DevOps – AWS, EKS, Docker, Kubernetes, Helm, CI/CD, Git, Github

Data & Operations – Data Analysis, Data Management, Structured and Unstructured Data, Production Systems, Software Algorithms, Debugging, Testing, Installation, Maintenance

PROJECTS

Autonomous Agentic Workflow for Data Reconciliation

- Developed a system of interconnected **Python AI agents** using **LangGraph** to automate complex financial data reconciliation tasks, processing approximately **5,000 transactions** per hour.
- Designed agent communication protocols enabling seamless **Agent-to-Agent (A2A)** interaction for task delegation, error reporting, and collaborative problem-solving, reducing manual review cycles by **35%**.
- Integrated the agent system with existing **PostgreSQL** databases and **Splunk** for logging and monitoring, ensuring data integrity and operational visibility.

Optimized LLM Inference Service for Real-time Analytics

- Built a low-latency inference service using **FastAPI** and **vLLM** to serve a custom-trained **machine learning model** for real-time anomaly detection on financial transactions.
- Achieved a p95 latency of **180ms** and throughput of **250 requests** per second by optimizing model quantization and parallel processing within the serving pipeline.
- Deployed the inference service as a **Docker** container on **AWS Lambda**, managing infrastructure and scaling through **CloudWatch** for monitoring and alerting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)